BUSS386 Problem Set 7 — Solutions

Properties of Options: Bounds and Parity

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Problem 1 — Six determinants

(a)

Factor (increase ↑)	Eur. c	Eur. p	Am. C	Am. P
Spot S_0	↑	↓	↑	↓
Strike K	↓	↑	↓	↑
Volatility σ	↑	↑	↑	↑
Dividends D	↓	↑	↓	↑
Risk-free rate r	↑	↓	↑	↓
Time to expiry T	**?**	**?**	↑	↑

(b) $\boxed{\text{Time to expiry}}$ is the row that differs. A longer-dated *American* option carries every exercise opportunity the shorter-dated one has, and more, so more time can never hurt: the direction is unambiguously ↑, by pure dominance and with no model needed. A *European* holder has only the single date T, so a longer wait cuts both ways — a large dividend may now fall inside the window, and on a put you simply receive K later, which is worth less today. The "?" therefore means *genuinely ambiguous*, not merely unproven.

Special case worth knowing: on a **non-dividend** stock the European call *is* unambiguously increasing in T, because it then coincides with the American call.

(c) Payoffs are kinked at K (capped at zero on the wrong side), so a wider terminal distribution adds upside without symmetric downside. Both calls and puts gain.

(d) An American put can be optimally exercised deep ITM: the interest earned on K exceeds the lost insurance value. An American call on a non-dividend stock has no such gain — holding it always dominates exercising, since you would forgo both time value and interest on K.

Problem 2 — Put-call parity bounds for American options

$Ke^{-rT} = 100\,e^{-0.03} = \$97.0446$.

(a) $S_0 - K \;\le\; C_0 - P_0 \;\le\; S_0 - Ke^{-rT}$, so $\boxed{0 \;\le\; C_0 - P_0 \;\le\; 2.9554}$.

(b) $C_0 - P_0 = 8.50 - 6.50 = 2.00$, which lies inside $[0,\ 2.9554]$. $\boxed{\text{No arbitrage}}$ — the bounds are satisfied.

(c) $C_0 - P_0 = 8.50 - 3.00 = 5.50 > 2.9554$: the **upper** bound is breached by $5.50 - 2.9554 = \boxed{\$2.5446}$. The call is rich relative to the put, so sell the synthetic and buy the real thing:

	Today	At T: $S_T \geq K$	At T: $S_T < K$
Sell the call	$+8.50$	$-(S_T - 100)$	0
Buy the put	-3.00	0	$+(100 - S_T)$
Buy the stock	-100.00	$+S_T$	$+S_T$
Borrow \$94.50	$+94.50$	-97.3780	-97.3780
Net	**0**	**+2.6220**	**+2.6220**

The loan grows to $94.50\,e^{0.03} = 97.3780$, and either state delivers the stock for $K = 100$. Riskless $\boxed{\$2.6220}$ at T, worth $2.6220\,e^{-0.03} = \$2.5446$ today — exactly the amount by which the bound was breached.

(d) If the call is assigned at some $t < T$, you hand over the stock and receive $K = 100$ immediately, while the loan has only grown to $94.50\,e^{0.06t} < 97.3780$. You bank at least $100 - 94.50\,e^{0.06t} > \$2.62$, *and* you still hold the put, whose value is ≥ 0. Early assignment therefore makes you $\boxed{\text{strictly better off}}$, never worse.

(e) Because the **put** is American. The lower bound is proved by showing that portfolio A (American call + bond) dominates portfolio B (American put + stock). But B can be cashed out at *any* moment: exercising the put at time t turns B into exactly K. So A has to be worth at least K at every such moment, which means its bond must start at the full $\boxed{K}$ today. Had A started with only Ke^{-rT}, at an early exercise date t it would hold just $Ke^{-rT}e^{rt} = Ke^{-r(T-t)} < K$ and the dominance would fail.

Paying for that worst case costs exactly the discounting, which is why the American bound is the weaker $S_0 - K$. The European put can only be exercised at T, so there is no early date to defend and Ke^{-rT} suffices.

Problem 3 — Put-call parity check on SPX

(a) Parity (European, no dividends): $\boxed{c_0 + Ke^{-rT} = p_0 + S_0}$.

(b) LHS $= 140 + 5{,}800 \cdot e^{-0.04 \cdot 0.25} = 140 + 5{,}800 \cdot 0.99005 = 140 + 5{,}742.3 = 5{,}882.3$.
RHS $= 80 + 5{,}800 = 5{,}880$.
Discrepancy $= +\$2.30$.

(c) LHS (call + bond) is rich by \$2.30 relative to RHS (put + stock). **Sell the call, buy the put, buy the stock, short the zero**. Cash flow today: $+140 - 80 - 5{,}800 + 5{,}742.3 = \boxed{+\$2.30}$. All terminal exposures cancel (both portfolios pay $\max(S_T, K)$). Whether this is exploitable depends on whether \$2.30 exceeds bid-ask + commissions; on SPX it typically does not.

Problem 4 — Lower-bound arbitrage on a European put

(a) Lower bound: $\max(Ke^{-rT} - S_0,\, 0) = \max(40e^{-0.025} - 37,\, 0) = \max(39.012 - 37,\, 0) = \boxed{\$2.012}$.
Quote \$1.00 $<$ \$2.012 $\Rightarrow$ arbitrage.

(b) Strategy: buy the put, buy the stock, borrow Ke^{-rT}.

Action today	CF now	CF at T	
		$S_T \geq 40$	$S_T < 40$
Buy put	-1.00	0	$40 - S_T$
Buy share	-37	S_T	S_T
Borrow $40e^{-0.025}$	$+39.012$	-40	-40
Net	$\boxed{+1.012}$	$S_T - 40 \geq 0$	0

(c) Risk-free profit today = $\boxed{\$1.012}$ per share, plus a non-negative upper branch at T.

Problem 5 — Forward-style parity on KOSPI 200

$T = 0.5$, $S_0 = 1{,}105$, $K = 1{,}105$, $r = 3\%$, $q = 1.5\%$, multiplier $M = ₩\,250{,}000$.

(a) $F_0 = S_0 e^{(r-q)T} = 1{,}105 \cdot e^{0.0075} = 1{,}105 \cdot 1.00753 = \boxed{1{,}113.32}$.

(b) Parity: $c_0 - p_0 = (F_0 - K)e^{-rT}$.
RHS $= (1{,}113.32 - 1{,}105)e^{-0.015} = 8.32 \cdot 0.98511 = \boxed{8.19}$ pts.
LHS $= 61.50 - 51.20 = 10.30$ pts.
Discrepancy $= 10.30 - 8.19 = \boxed{+2.11}$ pts. Call is rich relative to put.

(c) Sell the call, buy the put, buy e^{-qT} shares of the index, short the zero Ke^{-rT}. (Equivalently: sell the synthetic forward (long call + short put) at 4.30, buy the actual forward at 2.67.)

(d) Per contract, arbitrage $= 2.11 \cdot M = 2.11 \cdot 250{,}000 = \boxed{₩\,527{,}500}$ today (before bid-ask).

Problem 6 — Discrete-dividend parity

(a) $D = 1.50 \cdot e^{-0.04 \cdot 0.5} = 1.50 \cdot 0.9802 = \boxed{\$1.4703}$.

(b) Parity: $p_0 = c_0 + Ke^{-rT} - (S_0 - D) = 6 + 50e^{-0.04} - (52 - 1.4703) = 6 + 48.039 - 50.530 = \boxed{\$3.51}$.

(c) Doubling the dividend to \$3.00 doubles D to \$2.9405. Holding c_0 fixed and reapplying parity: $\Delta p = \Delta D = +1.4703 \Rightarrow$ new $p = 3.51 + 1.47 = \boxed{\$4.98}$ (an increase of \$1.47 per dollar of PV-dividends).

Problem 7 — Early exercise of American options

(a) **Result.** Never optimal to early-exercise an American call on a non-dividend stock; therefore $C_0 = c_0$.
Proof. For any $t < T$: $C_t \geq c_t = p_t + S_t - Ke^{-r(T-t)} = (S_t - K) + p_t + K[1 - e^{-r(T-t)}] > S_t - K$. Holding (or selling) the option dominates exercising.

(b) **Example.** $S_t = 100$, $K = 80$, $r = 4\%$ c.c., $T - t_d = 0.5$ yr remaining, large dividend $D = 5$, put-side time value $p_{t_d} \approx 0.50$. Heuristic threshold: $K(1 - e^{-r(T-t_d)}) + p_{t_d} = 80 \cdot (1 - 0.9802) + 0.50 = 1.584 + 0.50 = 2.08$. $D = 5 > 2.08 \Rightarrow$ exercise just before ex-dividend. $\boxed{\text{Optimal to exercise.}}$

(c) Exercising the put pays $K - S_t$ now; the cash K earns interest. When the put is deep ITM, the time value of waiting is small, and the interest gain on K dominates — exercising captures it.